



DATA ANALYTICS PROJECT

SUPERSTORE PERFORMANCE ANALYSIS

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BACKGROUND

- ❑ Superstore merupakan perusahaan yang bergerak di Amerika Serikat dimana menjual produk kebutuhan kantor, perabotan dan teknologi. Revenue perusahaan terus meningkat selama periode 2014-2017, namun tidak semua produk dan strategi penjualan memberikan kontribusi positif terhadap profit.
- ❑ Oleh karena itu, diperlukan analisis untuk memahami keseluruhan performa bisnis dari Superstore.

A woman with long dark hair, wearing a striped long-sleeved shirt and dark jeans, is pushing a metal shopping cart through an outdoor furniture store. She is walking away from the camera down a paved aisle. To her left are tall metal shelving units filled with boxes. To her right are various outdoor furniture items, including a large white patio umbrella, a table, and chairs. In the background, a sign on a building reads "SELF SERVE FURNITURE".

ASK PHASE

BUSINESS QUESTION

- ❑ Bagaimana trend perusahaan dari waktu ke waktu?
- ❑ Manakah region, kategori, dan segment yang memiliki performa bisnis paling kuat?

A woman with long dark hair, wearing a striped long-sleeved shirt and dark jeans, is seen from behind, pushing a metal shopping cart. She is walking through an outdoor furniture display area within a large warehouse. To her left are tall metal shelving units filled with boxes. To her right are various outdoor furniture items, including a large white patio umbrella, a table, and chairs. In the background, a sign on a wall reads "SELF SERVE FURNITURE". The entire scene is overlaid with a dark, semi-transparent banner containing the text "PREPARE PHASE" in large, white, bold, sans-serif capital letters. The text is underlined.

PREPARE PHASE

DATASET OVERVIEW

Analisis ini menggunakan dataset dari:

<https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

Total Row : 9984 row

Total Column : 21 column

A woman in a striped shirt and jeans is pushing a shopping cart through an outdoor furniture section of a warehouse store. The area is filled with various outdoor furniture items like tables, chairs, and umbrellas. In the background, there are high industrial shelving units and a sign that reads "SELF SERVE FURNITURE".

PROCESS PHASE

DATA PREPROCESSING:

01

MISSING VALUE

Tidak ada missing value pada seluruh column di dataset ini.

02

DUPLICATE VALUE

Ditemukan duplikat pada Row_ID 3406 dan 3407. Salah satu row dihapus untuk menjaga integritas analisis.

03

OUTLIER VALUE

Pengecekan pada kolom Sales dan Discount tidak menemukan nilai anomali yang dapat mengganggu hasil analisis.

A woman with long dark hair, wearing a striped long-sleeved shirt and dark jeans, is seen from behind, pushing a metal shopping cart. She is walking through an outdoor furniture section of a large warehouse-style store. To her left are tall metal shelving units filled with boxes. To her right are various outdoor furniture items, including a large white patio umbrella, a table, and chairs. In the background, a sign on a wall reads "SELF SERVE FURNITURE". The entire scene is overlaid with a dark horizontal band containing the text "ANALYZE PHASE" in white, bold, sans-serif font, which is underlined.

ANALYZE PHASE

```

SELECT
    SUM(SALES) AS 'Total Revenue',
    SUM(PROFIT) AS 'Total Profit',
    COUNT(DISTINCT Order_ID) AS 'Total Order',
    CASE
        WHEN SUM(Sales) = 0 THEN NULL
        ELSE
            (SUM(Profit)/SUM(Sales))*100 END AS 'Profit Margin'
FROM ss_master_table;

```

	Total Revenue	Total Profit	Total Order	Profit Margin
1	2296919.4883	286409.0805	5009	12.469200

Pada periode 2014-2017, telah didapatkan bahwa:

- Total Revenue : 2.296.919\$
- Total Profit : 286.409\$
- Total Order : 5009
- Profit Margin : 12.469%

```

SELECT
    YEAR(Order_Date) AS 'Year',
    CONCAT('Q', DATEPART(QUARTER, Order_Date)) AS Quarter,
    SUM(Sales) AS 'Total Revenue',
    SUM(Profit) AS 'Total Profit',
    COUNT(DISTINCT Order_ID) AS 'Total Order',
    CASE WHEN SUM(Sales) = 0 THEN NULL
    ELSE
        (SUM(Profit)/SUM(Sales))*100 END AS 'Profit Margin'

FROM ss_master_table
GROUP BY
    YEAR(Order_Date),
    DATEPART(QUARTER, Order_Date)
ORDER BY 'Year' DESC;

```

	Year	Quarter	Total Revenue	Total Profit	Total Order	Profit Margin
1	2017	Q1	123144.8602	23506.2026	240	19.088200
2	2017	Q2	133764.3720	15499.2085	367	11.586900
3	2017	Q3	196251.9560	26985.1325	448	13.750200
4	2017	Q4	280054.0670	27448.7260	632	9.801200
5	2016	Q1	93237.1810	11441.3708	179	12.271200
6	2016	Q2	136082.3010	16390.3394	294	12.044400
7	2016	Q3	143787.3622	15823.6048	378	11.004800
8	2016	Q4	236098.7538	38139.8593	464	16.154100
9	2015	Q1	68851.7386	9264.9416	144	13.456300
10	2015	Q2	89124.1870	12190.9224	214	13.678500
11	2015	Q3	130259.5752	16853.6194	274	12.938400
12	2015	Q4	182297.0082	23309.1203	406	12.786300
13	2014	Q1	74447.7960	3811.2290	131	5.119300
14	2014	Q2	86257.3876	11216.1280	201	13.003000
15	2014	Q3	143633.2123	12804.7218	267	8.914800
16	2014	Q4	179627.7302	21723.9541	370	12.093800

Berdasarkan trend dari 2014 hingga 2017, telah didapatkan bahwa

- Q4 memiliki revenue, profit, dan total order tertinggi. Q4 konsisten menjadi yang tertinggi selama periode tersebut.

- Q1 2017 memiliki profit margin tertinggi.

```

SELECT Region,
       SUM(SALES) AS 'Total Revenue',
       SUM(PROFIT) AS 'Total Profit',
       COUNT(DISTINCT Order_ID) AS 'Total Order',
       CASE WHEN SUM(Sales) = 0 THEN NULL ELSE (SUM(Profit)/SUM(Sales))*100 END AS 'Profit Margin'
FROM ss_master_table
GROUP BY Region;

```

	Region	Total Revenue	Total Profit	Total Order	Profit Margin
1	Central	501239.8908	39706.3625	1175	7.921600
2	East	678499.8680	91534.8388	1401	13.490700
3	South	391721.9050	46749.4303	822	11.934300
4	West	725457.8245	108418.4489	1611	14.944800

Berdasarkan analisa di atas, Region West memiliki performa terbaik karena memiliki total revenue, total profit, total order, dan profit margin tertinggi.

Berdasarkan analisa di bawah, Technology menjadi kategori yang memiliki performa terbaik jika dilihat dari keempat KPI.

```
SELECT Category,  
       SUM(SALES) AS 'Total Revenue',  
       SUM(PROFIT) AS 'Total Profit',  
       COUNT(DISTINCT Order_ID) AS 'Total Order',  
       CASE WHEN SUM(Sales) = 0 THEN NULL ELSE (SUM(Profit)/SUM(Sales))*100 END AS 'Profit Margin'  
FROM ss_master_table  
GROUP BY Category;
```

	Category	Total Revenue	Total Profit	Total Order	Profit Margin
1	Furniture	741718.4233	18463.3316	1764	2.489200
2	Office Supplies	719047.0320	122490.8008	3742	17.035100
3	Technology	836154.0330	145454.9481	1544	17.395700

- Technology menjadi kategori dengan total revenue, total profit, dan profit margin tertinggi.
- Office Supplce menjadi kategori dengan total order tertinggi.

- Consumer menjadi segmen dengan total revenue, total profit, dan total order tertinggi.
- Home Office menjadi segmen dengan profit margin tertinggi.

```
SELECT Segment,  
       SUM(SALES) AS 'Total Revenue',  
       SUM(PROFIT) AS 'Total Profit',  
       COUNT(DISTINCT Order_ID) AS 'Total Order',  
       CASE WHEN SUM(Sales) = 0 THEN NULL ELSE (SUM(Profit)/SUM(Sales))*100 END AS 'Profit Margin'  
FROM ss_master_table  
GROUP BY Segment;
```

	Segment	Total Revenue	Total Profit	Total Order	Profit Margin
1	Consumer	1161401.3450	134119.2092	2586	11.548000
2	Corporate	706146.3668	91979.1340	1514	13.025500
3	Home Office	429371.7765	60310.7373	909	14.046200

Berdasarkan analisa di di atas, Consumer menjadi segmen yang memiliki performa terbaik jika dilihat dari keempat KPI.

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SHARE PHASE



Year

All

Quarter

All

Category

All

Reset Filter

Performance Overview 2014-2017

Revenue

Profit

Orders

Profit Margin

\$2.30M

Total Revenue

\$286.41K

Total Profit

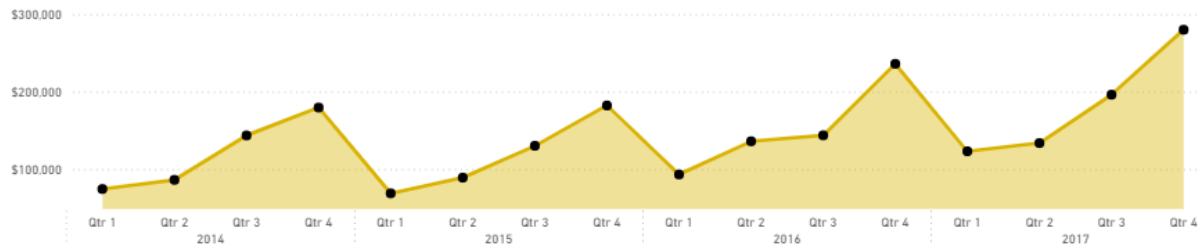
5009

Total Order

12.47%

Profit Margin

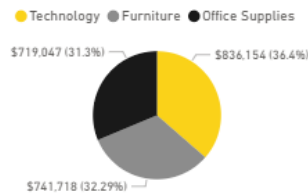
Revenue Trend Over Time



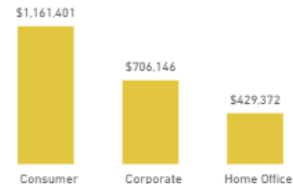
Revenue by Region



Revenue by Category



Revenue by Segment



DASHBOARD

A woman with long dark hair, wearing a striped long-sleeved shirt and dark jeans, is pushing a shopping cart through an outdoor furniture store. She is walking away from the camera down a paved aisle. To her left are tall metal shelving units filled with boxes. To her right are various outdoor furniture items, including a large white patio umbrella, a table, and chairs. In the background, a sign on a building reads "SELF SERVE FURNITURE".

ACT PHASE

RECOMMENDATION

- ❑ Tingkatkan inventory dan aktivitas marketing menjelang Q4.
- ❑ Implementasi cross-selling untuk meningkatkan performa dari segmen Corporate dan Home Office.
- ❑ Prioritaskan investasi di region West karena memiliki peluang mendapatkan revenue dan profit yang lebih tinggi.
- ❑ Prioritaskan investasi pada kategori Technology sebagai profit driver utama.